

Discovering the Fourier Basis and the Multiplication Law of Convolution from Data

Name

Abstract

Convolution in the time domain is equivalent to multiplication in the frequency domain. This report treats that theorem as a discovery problem: given random triples (x, h, y) with $y = h \otimes x$, can a learner recover both the hidden basis and the coordinate law without being shown the DFT or multiplication? The project uses an AI-assisted workflow: Claude helped generate the idea and coding plan, while Codex implemented the experiments, revised the loss after an initial failure mode, and ran the final sweeps. We train a spectral model with a free complex basis A and a shared coordinate-wise MLP g_ϕ . Theory shows that any basis diagonalizing all circulant matrices must be the DFT up to row permutation and row scaling, and that the coordinate law is then complex multiplication up to scale. Empirically, the full model recovers this structure for $n = 16, 32, 64$, with diagonalization residuals around 10^{-5} and multiplication-probe residuals around 10^{-3} . The ablation shows that MSE-only lowers prediction error without recovering a clean Fourier basis or multiplication law, while diagonalization-only recovers the basis but leaves the coordinate law untrained.

Keywords

Fourier Transform, Representation Learning, AI for Mathematics

1 Introduction

Convolution is a basic operation in signal processing, time-series analysis, and convolutional neural networks. Its main mathematical simplification is the convolution theorem:

$$F(h \otimes x) = (Fh) \odot (Fx), \quad (1)$$

where F is the DFT matrix and $\odot$ denotes entrywise multiplication. Classical texts use this result in the forward direction: the Fourier basis and the product law are already known, so convolution can be analyzed and computed efficiently [1–3]. In practice, this is why linear filters are often described by their frequency response: a moving average suppresses high-frequency variation, a band-pass filter keeps a selected frequency range, and a convolutional filter can be interpreted by how it scales different frequency components. The theorem is therefore not only a computational shortcut, but also a statement about the coordinate system in which convolution has its simplest form.

This project asks the inverse question. Suppose a model only sees random examples of circular convolution and is told one weak structural prior: in some unknown linear basis, convolution should act independently on each coordinate. Is the Fourier basis then forced by the data? This is a small but verifiable AI-for-mathematics experiment, in the same spirit as using computation to guide or test mathematical patterns [6, 7]. The setting is useful because the correct answer is known exactly. Unlike an open-ended pattern search, we can check whether the learned object is actually the DFT,

whether the learned coordinate operation is actually multiplication, and which training ingredients are needed for the rediscovery.

The experiment also separates prediction from mathematical discovery. A flexible model may achieve a moderate test error while representing convolution in an opaque way. That is not enough for this assignment, whose goal is to state and test a mathematical pattern. Therefore the evidence below measures not only output error, but also the algebraic structure of the learned basis and the learned coordinate law. In this sense, the project treats a classical theorem as a target for rediscovery: the theorem predicts a unique hidden representation, and the data-science question is whether a learner can recover it from examples.

The mathematical pattern tested here is the following.

CONJECTURE 1.1 (REDISCOVERY). *Under the prior that a fixed linear basis turns circular convolution into a shared coordinate-wise interaction, gradient-based learning recovers the DFT basis (up to permutation, phase, and scale) and the coordinate law of complex multiplication (up to one global complex scale).*

The report keeps two goals separate. First, Section 3 explains why the conjecture is mathematically well-posed: simultaneous diagonalization of all circulant matrices uniquely characterizes the DFT. Second, Sections 4–5 test whether a learner with a free basis and a shared coordinate MLP can find this structure in practice. Additional proof details, large tables, and figures are placed in the appendix to keep the main text focused while preserving the evidence.

2 AI-Assisted Development Workflow

This project was developed through a human-in-the-loop AI workflow. I first used Claude to brainstorm AI-for-math ideas. The original convolution idea was narrower than the final project: it effectively assumed the coordinate-wise product was already known and only tried to learn an FFT-like matrix. I manually changed the research question so that both unknown objects had to be discovered: the hidden basis A and the shared coordinate law g_ϕ . This made the task closer to the assignment goal, because the learner had to recover not only the Fourier basis but also the multiplication rule. After that, I asked Claude to produce a coding plan covering data generation, model design, training, metrics, and experiments, and I copied that plan into the project Markdown file.

Codex was then used for implementation and iteration. In plan mode, it translated the Markdown plan into code for direct circular convolution, the complex basis model, the coordinate MLP, and the evaluation metrics. The first implementation mainly used MSE loss, and the results were not satisfactory: prediction error could decrease, but the learned basis was not clearly Fourier-like and the learned law was not multiplication. I then used Codex’s seek-goal mode with the explicit goal of exposing the Fourier structure. Codex added the diagonalization loss, row-norm gauge fix, and three-phase

training schedule, and ran the main size sweep, ablation, robustness, and baseline experiments. After the presentation, I added further experiments based on feedback, especially clearer ablations and extra robustness or control results. Thus the final project was not a one-shot AI output; it was an iterative process in which AI tools proposed, implemented, and revised the experiment, while the human role was to refine the mathematical target and interpret the failures.

3 Mathematical Background

All indices are modulo n , and $\omega = e^{-2\pi i/n}$. For $x, h \in \mathbb{R}^n$, circular convolution is

$$(h \otimes x)_t = \sum_{s=0}^{n-1} x_s h_{(t-s) \bmod n}. \quad (2)$$

For fixed h , this is a linear map $x \mapsto C_h x$, where C_h is circulant:

$$(C_h)_{t,s} = h_{(t-s) \bmod n}. \quad (3)$$

If S is the cyclic shift matrix, then every circulant matrix can be written as

$$C_h = \sum_{r=0}^{n-1} h_r S^r. \quad (4)$$

Thus all circulants commute and are generated by one shift operator. This simple algebraic fact is the reason the discovery problem is identifiable. A generic family of matrices would not share an eigenbasis, but all convolution operators are polynomials of the same shift. If a learner finds a basis that diagonalizes many random convolution matrices, it is implicitly learning the eigenbasis of S .

For intuition, when $n = 4$ and $h = (h_0, h_1, h_2, h_3)$,

$$C_h = \begin{pmatrix} h_0 & h_3 & h_2 & h_1 \\ h_1 & h_0 & h_3 & h_2 \\ h_2 & h_1 & h_0 & h_3 \\ h_3 & h_2 & h_1 & h_0 \end{pmatrix}.$$

A dense kernel gives a dense circulant matrix; a sparse kernel gives a banded circulant with wrap-around entries. Both are governed by the same shift matrix, so a basis discovered from dense Gaussian kernels should also work on sparse and structured kernels if it has learned convolution itself rather than a training distribution artifact.

LEMMA 3.1 (DFT DIAGONALIZATION). *Let $F_{m,t} = \omega^{mt}$ be the DFT matrix. For every kernel h ,*

$$FC_h F^{-1} = \text{diag}(Fh). \quad (5)$$

Equivalently, convolution by h acts in Fourier coordinates as $Fx \mapsto (Fh) \odot (Fx)$.

The key fact for discovery is that the DFT is not merely one useful basis; it is the only basis with this decoupling property, modulo unavoidable symmetries. The symmetries are important. Permuting Fourier rows only relabels frequencies, and scaling a row only changes the units of that coordinate. No data-driven method can identify these choices uniquely, so the evaluation below explicitly quotients them out.

THEOREM 3.2 (UNIQUENESS OF THE DIAGONALIZING BASIS). *Let $A \in \mathbb{C}^{n \times n}$ be invertible. Then*

$$AC_h A^{-1} \text{ is diagonal for every } h \iff A = PDF,$$

where P is a permutation matrix and D is an invertible diagonal matrix.

PROOF SKETCH. The forward direction follows from Lemma 3.1: multiplying F on the left by a row permutation or a nonzero row scaling preserves diagonality. For the reverse direction, it is enough to diagonalize the shift S , because every circulant is a polynomial in S by (4). The eigenvalues of S are the distinct n -th roots of unity, so every eigenspace is one-dimensional. Hence the columns of A^{-1} must be the Fourier eigenvectors, in some order and with nonzero scalings. Therefore $A = PDF$. Full details are in Appendix A. $\square$

COROLLARY 3.3 (COORDINATE LAW). *After fixing the representative $A = F$, any shared coordinate-wise map satisfying $A^{-1}g(Ax, Ah) = h \otimes x$ for all x, h must be $g(u, v)_k = u_k v_k$, up to one global nonzero complex scale.*

The theorem and corollary justify the experiment: if the learner truly finds a basis in which all convolution operators decouple, then the learned basis should match the DFT and the learned scalar law should match multiplication. They also explain why the result is stronger than simply fitting examples. A model that does not produce a joint diagonalizer has not recovered the theorem, even if its output MSE is acceptable.

4 Method

The model contains two learnable objects: an unconstrained complex matrix $A \in \mathbb{C}^{n \times n}$ and a shared scalar MLP $g_\phi : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$. For each pair (x, h) ,

$$u = Ax, \quad v = Ah, \quad z_k = g_\phi(u_k, v_k), \quad \hat{y} = A^{-1}z. \quad (6)$$

In real-valued implementation, the MLP input is $[\Re u_k, \Im u_k, \Re v_k, \Im v_k]$ and the output is $[\Re z_k, \Im z_k]$. The same A encodes both x and h ; this tied design fixes the relative gauge between the two inputs and makes the multiplication law easier to identify. Importantly, A is not parameterized as a Fourier matrix, and no FFT or built-in convolution is used. The direct convolution formula is used only to generate labels, so the learner is not given either side of the conclusion in Eq. (1).

The shared MLP is deliberately restrictive. It receives only the pair (u_k, v_k) from one coordinate and must use the same parameters for every k . Thus it cannot hide a different rule in each frequency coordinate. If the basis is correct, the same scalar operation should work everywhere; if the basis is wrong, the missing cross-coordinate interactions must be absorbed by a function that is not allowed to see them. This design turns the mathematical statement “convolution decouples in Fourier coordinates” into a testable learning bias.

The only structural assumption is the following.

Assumption 1 (Coordinate-decoupling prior). *There exists an invertible linear basis A in which every convolution operator acts coordinate-wise, i.e., $AC_h A^{-1}$ is diagonal for every kernel h .*

Training uses two losses. The output loss matches the generated convolution:

$$\mathcal{L}_{\text{out}} = \frac{1}{n} \|\hat{y} - y\|^2. \quad (7)$$

The diagonalization penalty encodes Assumption 1 without referencing the DFT:

$$\mathcal{L}_{\text{diag}} = \frac{\sum_{i \neq j} |(AC_h A^{-1})_{ij}|^2}{\sum_{i,j} |(AC_h A^{-1})_{ij}|^2}. \quad (8)$$

This term says only that the learned basis should decouple the operator; by Theorem 3.2, the minimizers of this structural condition are exactly the Fourier family *PDF*.

After each optimization step, every row of A is normalized to unit norm. This removes the unidentifiable row-scale freedom in Theorem 3.2; it does not impose any Fourier row or multiplication formula. Without this gauge fix, row norms can drift, and the single shared MLP must learn the same multiplication law under inconsistent input scales. The normalization only chooses a representative inside the equivalence class *PDF*.

The training schedule has three phases. Phase 1 trains A and g_ϕ using $\mathcal{L}_{\text{out}} + \lambda \mathcal{L}_{\text{diag}}$, so the basis is pushed toward a decoupled coordinate system. Phase 2 freezes A and trains only g_ϕ on the output loss, allowing the coordinate law to stabilize in a fixed coordinate frame. Phase 3 fine-tunes g_ϕ at a smaller learning rate. This separation is useful because the basis and law otherwise chase each other: a changing basis changes the input distribution seen by g_ϕ , while an untrained g_ϕ gives a noisy signal for how A should move.

5 Evaluation

Each training example is generated by drawing $x \sim \mathcal{N}(0, I)$ and $h \sim \mathcal{N}(0, I)$, then computing $y = h \otimes x$ with the direct formula (2). Gaussian samples are used because they excite all coordinates and make the operator hard to fit by memorizing a few special cases. Two out-of-distribution kernel families are used only for testing: sparse kernels and structured kernels with a decaying envelope. For robustness, Gaussian noise is also added to training labels in a separate sweep, while test labels remain clean.

The evaluation measures four aspects. Test MSE checks task accuracy, but it is not sufficient by itself. The diagonalization residual is the off-diagonal energy fraction

$$\frac{\sum_{i \neq j} |(AC_h A^{-1})_{ij}|^2}{\sum_{i,j} |(AC_h A^{-1})_{ij}|^2},$$

which is near zero only when the learned basis jointly diagonalizes convolution operators. The DFT-row residual matches each row of A to the closest DFT row, allowing the row permutation, phase, and scale freedoms from Theorem 3.2. Finally, the multiplication probe fits a single complex scalar c and reports the relative error between $g_\phi(u, v)$ and $c(u \odot v)$ on encoded samples. This last metric asks whether the learned MLP behaves like one shared complex multiplication law, not merely whether the final output is close.

Table 1 summarizes the main size sweep. The learned basis nearly diagonalizes circulants, the rows align with DFT rows, and the coordinate law is very close to multiplication. The sparse and structured OOD errors are also small, which indicates that the model learned the convolution operator rather than only the Gaussian training distribution. The $n = 64$ result is particularly useful: although the

Table 1: Main results for the full objective. Lower is better.

n	Test	Sparse	Struct.	Diag.	Row	Probe
16	1.61e-3	3.97e-5	5.52e-5	1.55e-6	8.42e-4	2.39e-3
32	1.02e-3	2.83e-4	3.48e-4	1.07e-5	2.25e-3	3.38e-3
64	1.09e-3	1.48e-4	2.28e-4	8.16e-6	2.00e-3	1.54e-3

Table 2: Ablation at $n = 32$. Lower is better.

Objective	Test MSE	Diag. res.	Mult. probe
MSE + diag	1.0×10^{-3}	1.1×10^{-5}	3.4×10^{-3}
MSE only	3.9×10^{-2}	0.49	1.55
diag only	31.9	3.4×10^{-7}	24.4

space is larger and optimization is harder, the DFT-row residual and multiplication probe remain on the same order as the smaller cases.

The ablation in Table 2 shows why both losses are needed. With the diagonalization loss alone, A becomes Fourier-like, but g_ϕ is not trained to perform the task, so the multiplication probe and test MSE are poor. With the output loss alone, the model can reduce test error partly, but the basis does not become a clean joint diagonalizer and the learned coordinate law is not multiplication. In contrast, the full objective simultaneously gives low test MSE, near-zero diagonalization residual, and a small multiplication probe.

This pattern matches the theory. The diagonalization penalty identifies the coordinate system but does not tell the MLP which coordinate-wise operation to perform. The output loss teaches the operation but, without the diagonalization bias, can use less interpretable routes that do not expose the Fourier basis. The successful model is therefore not just a predictor; it is a predictor constrained to express the solution in the mathematically correct coordinates.

The noise and baseline studies, reported in Appendix B, give two additional checks. Under label noise, the diagonalization residual degrades more slowly than the multiplication probe, suggesting that the global basis is more stable than the learned scalar law. The oracle baselines confirm that the task and implementation are correct, while unstructured MLP controls have many more parameters but worse error and no interpretable coordinate law. These comparisons support the main claim that structure, rather than parameter count, is what makes the theorem rediscoverable.

6 Conclusion

This project framed the convolution theorem as a data-driven discovery task. The mathematical result shows that the coordinate-decoupling prior uniquely forces the DFT basis, up to row permutation and scaling, and then forces the coordinate-wise law to be complex multiplication. The empirical results support Conjecture 1.1: a free complex basis plus a shared coordinate MLP recovers both objects for $n = 16, 32, 64$ without hard-coding the DFT, FFT, or multiplication. The result is strongest when read together with the ablation: low prediction error alone is not the discovered pattern; the pattern is the joint recovery of the Fourier basis and the product law.

References

- [1] J. W. Cooley and J. W. Tukey. An algorithm for the machine calculation of complex Fourier series. *Mathematics of Computation*, 19(90):297–301, 1965.
- [2] R. N. Bracewell. *The Fourier Transform and Its Applications*. McGraw-Hill, 3rd edition, 2000.
- [3] A. V. Oppenheim, R. W. Schaffer, and J. R. Buck. *Discrete-Time Signal Processing*. Prentice Hall, 2nd edition, 1999.
- [4] S. Mallat. *A Wavelet Tour of Signal Processing: The Sparse Way*. Academic Press, 3rd edition, 2009.
- [5] M. M. Bronstein, J. Bruna, T. Cohen, and P. Velickovic. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.
- [6] A. Davies, P. Velickovic, L. Buesing, S. Blackwell, D. Zheng, N. Tomasev, R. Tanburn, P. Battaglia, C. Blundell, A. Juhasz, M. Lackey, G. Williamson, D. Hassabis, and P. Kohli. Advancing mathematics by guiding human intuition with AI. *Nature*, 600:70–74, 2021.
- [7] B. Romera-Paredes, M. Barekatin, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. S. Ellenberg, P. Wang, O. Fawzi, P. Kohli, and A. Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625:468–475, 2024.

A Proof Details

DETAILED PROOF OF LEMMA 3.1. Let w_m be the Fourier vector with $(w_m)_t = \omega^{-mt}$. Then

$$\begin{aligned} (C_h w_m)_t &= \sum_s h_{(t-s)} \omega^{-ms} = \sum_r h_r \omega^{-m(t-r)} \\ &= \omega^{-mt} \sum_r h_r \omega^{mr} = (Fh)_m (w_m)_t. \end{aligned}$$

Thus w_m is an eigenvector of C_h with eigenvalue $(Fh)_m$. Stacking these eigen-relations gives $FC_h F^{-1} = \text{diag}(Fh)$. $\square$

DETAILED PROOF OF THEOREM 3.2. If $A = PDF$, then

$$(PDF)C_h(PDF)^{-1} = PD(FC_h F^{-1})D^{-1}P^{-1} = P \text{diag}(Fh)P^{-1},$$

which is diagonal.

Conversely, suppose $AC_h A^{-1}$ is diagonal for every h . In particular, ASA^{-1} is diagonal. Since $S^n = I$, the eigenvalues of S are the n distinct roots of unity. Therefore each eigenspace is one-dimensional. The Fourier vectors form these eigenspaces by Lemma 3.1. Since ASA^{-1} is diagonal, the columns of A^{-1} must be eigenvectors of S . They must therefore be the Fourier eigenvectors in some order and with nonzero scalings, so $A^{-1} = F^{-1}\Pi\Delta$ for a permutation Π and invertible diagonal Δ . Equivalently, $A = PDF$. $\square$

DETAILED PROOF OF COROLLARY 3.3. With $A = F$, Lemma 3.1 gives

$$F(h \otimes x) = (Fh) \odot (Fx).$$

Writing $u = Fx$, $v = Fh$, and $z = F(h \otimes x)$, each coordinate satisfies $z_k = v_k u_k$. Hence any shared coordinate function reproducing convolution for all dense encoded inputs must agree with complex multiplication on its domain. The remaining scale freedom is a single global complex constant after row norms are fixed. $\square$

B Additional Results

Interpretation of the noise sweep. Table 3 studies whether the discovered structure is stable when the training labels are corrupted by Gaussian noise. As the SNR decreases from 40 dB to 0 dB, the clean test MSE increases from 1.58×10^{-3} to 1.21×10^{-1} , so the prediction task does become harder under noisy supervision. However, the structural metrics degrade much more slowly. The diagonalization residual remains small even at 0 dB (1.32×10^{-3}), meaning

that the learned basis still approximately diagonalizes the circulant operators. The DFT-row residual also increases gradually, from 1.76×10^{-3} to 2.57×10^{-2} , but it remains far from a random or unstructured basis. This suggests that the global Fourier basis is a robust and strongly identified part of the problem.

The multiplication probe is more sensitive to noise. It rises from 2.62×10^{-3} at 40 dB to 3.83×10^{-2} at 0 dB, a larger relative change than the diagonalization residual. This is consistent with the roles of the two learned objects. The basis A is constrained by many convolution operators at once, while the scalar law g_ϕ is trained directly from noisy output values. Therefore, noise first affects the local coordinate-wise map, and only later damages the global basis. Overall, the noise sweep supports the claim that the Fourier basis is not a fragile artifact of clean data: even under substantial label noise, the model still discovers a nearly diagonalizing, Fourier-aligned coordinate system.

Interpretation of the baselines and controls. Table 4 compares the proposed model with several controls at $n = 32$. The two oracle models give near-zero error: `HardProductOracle` uses the true DFT together with the exact complex product, while `HardProductLearnedBasis` uses the learned basis but replaces the learned MLP law by the hard-coded product. Their errors, 5.48×10^{-10} and 1.03×10^{-11} , verify that the data generation, convolution implementation, and evaluation pipeline are correct. More importantly, the second oracle shows that the learned basis is already accurate enough to support almost exact convolution once the correct coordinate law is supplied. Thus, the remaining gap of the full spectral model mainly comes from learning g_ϕ , not from failing to find the Fourier basis.

Among the non-oracle models, the proposed spectral model performs best, with Test MSE 1.02×10^{-3} . The BilinearTensor baseline is the closest competitor (3.44×10^{-2}), because convolution is a bilinear operation and a large bilinear tensor can partially fit it. However, this baseline does not reveal the Fourier basis or a simple coordinate-wise multiplication rule; it hides the structure inside an unstructured parameter tensor. The MLPControl has many parameters but performs worse (2.49×10^{-1}), showing that generic nonlinear approximation is not sufficient for clean discovery. The NonlinearEncoderControl fails almost completely, with error 8.25, indicating that a flexible encoder without the correct algebraic prior does not automatically find the convolution theorem. These comparisons show that the success of the spectral model is not merely due to model capacity. It comes from the combination of a linear basis, a shared coordinate-wise law, and the diagonalization objective, which together match the algebraic structure of circular convolution.

Table 3: Noise sweep, $n = 32$, full objective.

SNR (dB)	Test MSE	Diag. res.	DFT-row res.	Mult. probe
40	1.58×10^{-3}	6.19×10^{-6}	1.76×10^{-3}	2.62×10^{-3}
30	3.17×10^{-3}	2.13×10^{-5}	3.25×10^{-3}	7.81×10^{-3}
20	6.66×10^{-3}	7.25×10^{-5}	6.06×10^{-3}	7.60×10^{-3}
10	2.30×10^{-2}	3.03×10^{-4}	1.23×10^{-2}	1.31×10^{-2}
0	1.21×10^{-1}	1.32×10^{-3}	2.57×10^{-2}	3.83×10^{-2}

Table 4: Baselines and controls at $n = 32$.

Model	Test MSE
HardProductOracle	5.48×10^{-10}
HardProductLearnedBasis	1.03×10^{-11}
Spectral (ours)	1.02×10^{-3}
BilinearTensor	3.44×10^{-2}
MLPControl	2.49×10^{-1}
NonlinearEncoderControl	8.25

C Additional Figures

Interpretation of the learned basis. Figure 1 gives direct visual evidence that the full model recovers a Fourier-like basis, rather than merely fitting the output labels through an uninterpretable representation. The heatmap of $|A|$ is nearly constant around $1/\sqrt{32}$, which is exactly the magnitude pattern expected from a row-normalized DFT matrix. This is important because a generic learned complex matrix would not have equal-magnitude entries; it would typically show large variation across rows and columns. Therefore, the heatmap already suggests that the learned basis has the same global structure as the Fourier basis.

The row-alignment plots provide a more detailed check. After matching each learned row of A to its closest DFT row and allowing the unavoidable complex scaling freedom, the magnitude curves are almost flat and close to the DFT reference. The phase curves also nearly overlap with the corresponding DFT rows. This shows that the model is not only learning a matrix with the correct magnitude pattern, but also recovering the oscillatory phase structure that characterizes Fourier modes. Some phase plots contain sharp jumps near $\pm\pi$; these are caused by the wrap-around convention of complex arguments, where π and $-\pi$ represent the same direction. Thus, these discontinuities are not actual failures of alignment. Together, the heatmap and row plots support the claim that the learned basis is the DFT up to row permutation, phase, and scale.

Interpretation of the coordinate law and noise robustness. Figure 2 checks the second part of the discovery problem: whether the learned coordinate-wise map g_ϕ behaves like complex multiplication. Since the basis is identifiable only up to a global scale, the probe first fits one complex scalar c and then compares $g_\phi(u, v)$ with cuv . In the scatter plots, the real part of $g_\phi(u, v)$ is plotted against the real part of cuv , and the imaginary part is plotted against the imaginary part of cuv . The points fall almost exactly on the diagonal in both plots, with only small deviations at large magnitudes. This indicates that the MLP has learned the same algebraic law as Fourier-domain multiplication, rather than some unrelated nonlinear rule.

The noise-degradation plot adds a robustness check. When the SNR is high, both the diagonalization residual and the multiplication-probe residual are small, which means that the model discovers both the Fourier basis and the product law from nearly clean supervision. As the SNR decreases, both residuals increase, but they do not increase at the same rate. The diagonalization residual stays close to zero for a wide range of noise levels, while the multiplication-probe grows more noticeably. This difference is meaningful. The basis A is constrained by the global algebraic structure shared

across many random convolution operators, so it remains relatively stable under noisy labels. In contrast, the local coordinate map g_ϕ is trained directly from the noisy target outputs, and therefore absorbs more of the label corruption. This suggests that the Fourier basis is the more robust part of the rediscovered theorem, while the learned multiplication law is more sensitive to the quality of the supervision.

Interpretation of the diagonalization-only ablation. Figure 3 isolates the effect of the diagonalization penalty. The heatmap still has an almost uniform magnitude pattern, and the aligned rows match the DFT rows very closely in both magnitude and phase. This confirms the mathematical role of the diagonalization loss: by forcing AC_hA^{-1} to be diagonal for random circulant matrices C_h , it pushes the learned basis toward the common eigenbasis of all circulants, which is the Fourier basis. In this sense, the diagonalization penalty is sufficient to discover the correct coordinate system.

However, the same figure also shows why this objective alone is not sufficient for the full task. The multiplication-probe scatter plots collapse into a small, unstructured cloud and do not align with the diagonal. The reported residual is very large, showing that $g_\phi(u, v)$ is not approximating cuv . This failure is expected because the diagonalization-only objective never compares the predicted output $\hat{y}$ with the true convolution output y . It trains the basis to decouple the linear convolution operators, but it gives no learning signal that the coordinate-wise operation should be multiplication. Therefore, this ablation discovers the “where” part of the theorem—the Fourier coordinate system—but not the “what” part—the multiplication law inside each coordinate.

The noise plot for the diagonalization-only setting should be interpreted in the same way. Since the output law is not trained, the multiplication-probe residual remains large and unstable. The diagonalization residual, on the other hand, remains small because the basis objective is still directly optimized. This separation is useful evidence for the design of the full method: diagonalization is a powerful structural prior, but it must be paired with an output loss in order to learn the actual convolution map.

Interpretation of the MSE-only ablation. Figure 4 shows the opposite failure mode. With only the output MSE loss, the model receives a direct signal to predict $y = h \otimes x$, so it can partially reduce prediction error. However, the learned representation is no longer cleanly Fourier. The heatmap of $|A|$ is much less uniform than in the full model, and several rows show strong magnitude variation. This means that the learned basis does not have the equal-magnitude structure of a normalized DFT matrix. The row-alignment plots confirm the same point: some rows align only weakly with their matched DFT rows, and the phase curves often deviate from the DFT reference.

The multiplication-probe plots also show that the coordinate-wise law is not cleanly recovered. Although the scatter has some weak correlation with cuv , it is much more dispersed than in the full model, and the residual remains far larger. This means that the model is not representing convolution as simple Fourier-domain multiplication. Instead, it appears to reduce MSE through a more opaque representation in which the basis and the MLP compensate for each other. Such a solution may be useful for prediction, but it is not a mathematical rediscovery of the convolution theorem.

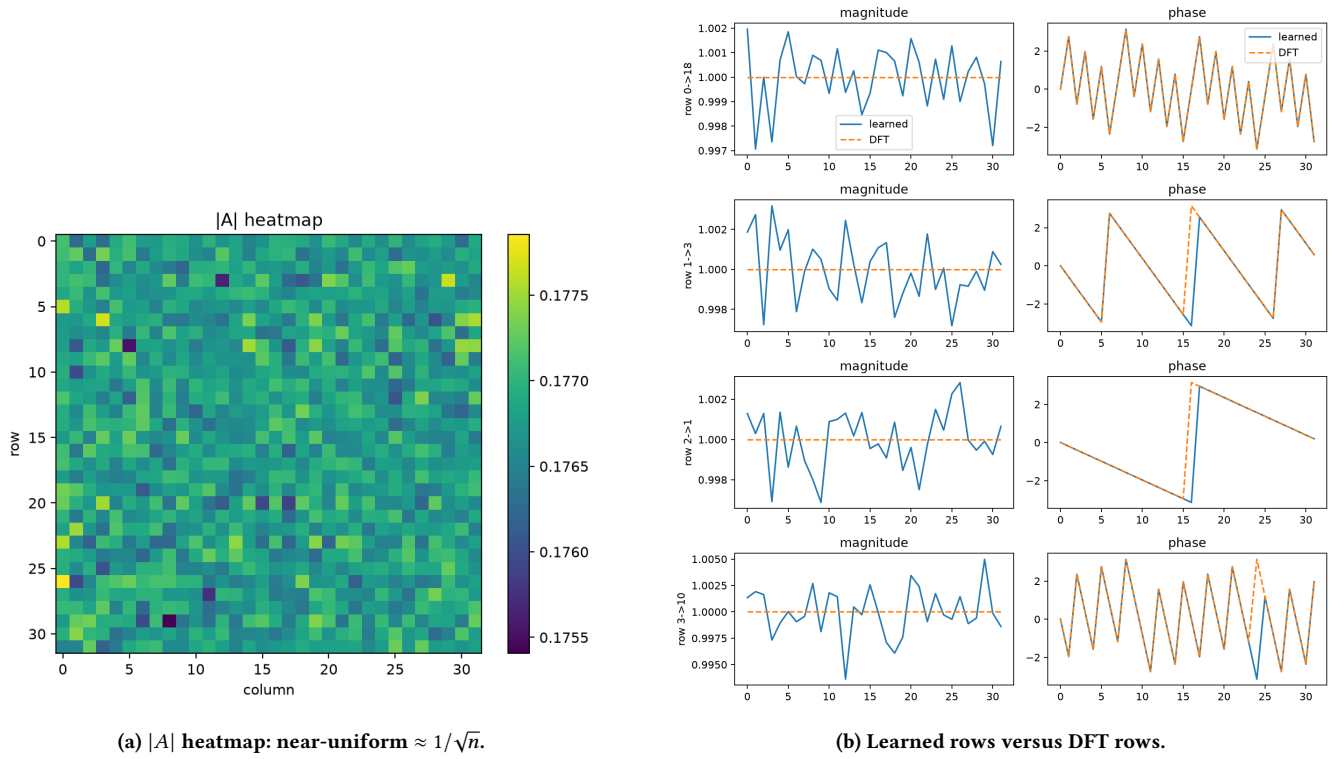


Figure 1: Basis diagnostics for the full model.

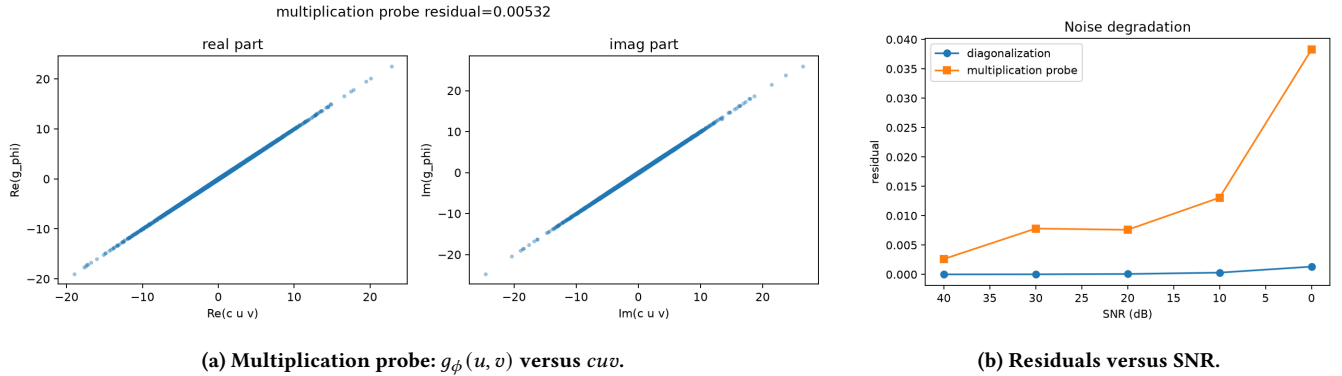


Figure 2: Law and robustness diagnostics for the full model.

This ablation is important because it shows that low output error alone is not enough for interpretable discovery. A flexible model can sometimes fit the task without exposing the underlying algebraic structure. The diagonalization loss is therefore not just a regularizer for better numerical performance; it is the component that makes the discovered solution mathematically meaningful. Comparing

the full model with the two ablations gives a clear division of labor: the diagonalization loss identifies the Fourier coordinate system, while the MSE loss trains the coordinate-wise operation to match convolution. Only their combination recovers both parts of the theorem.

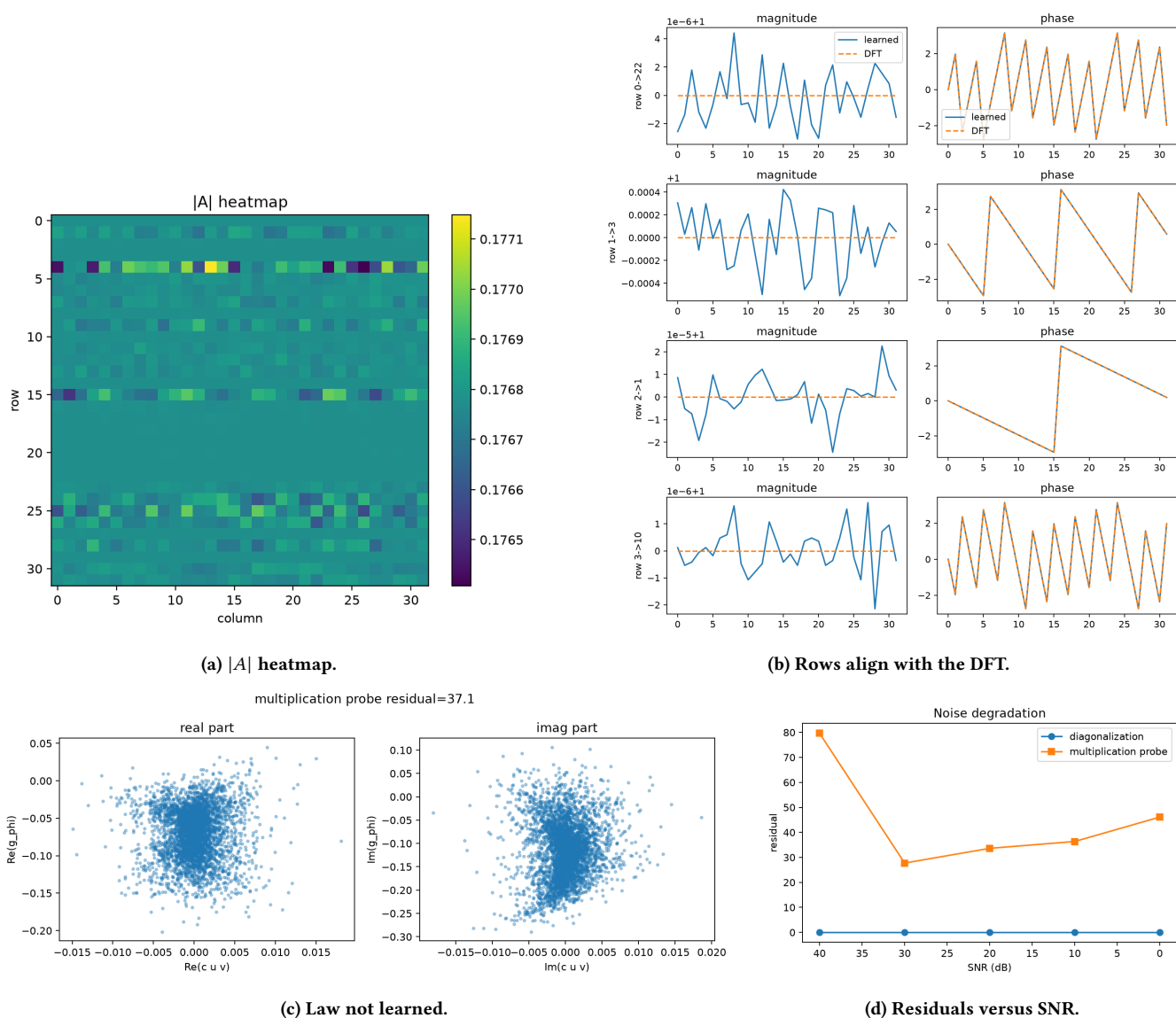
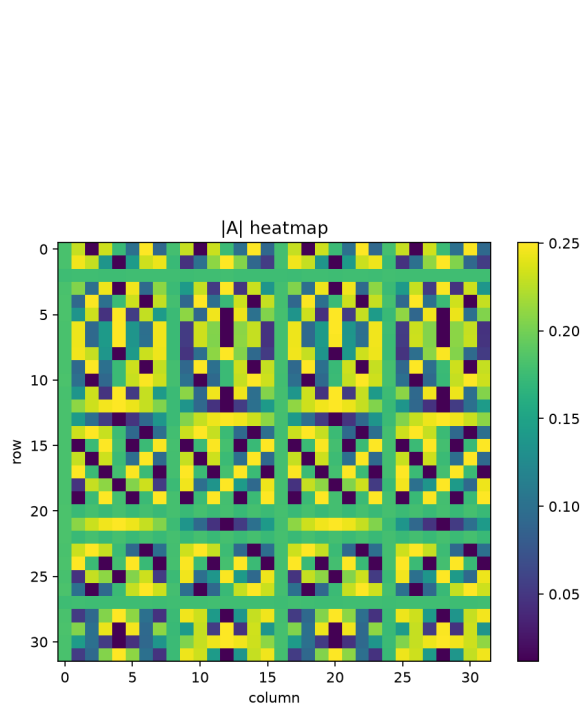
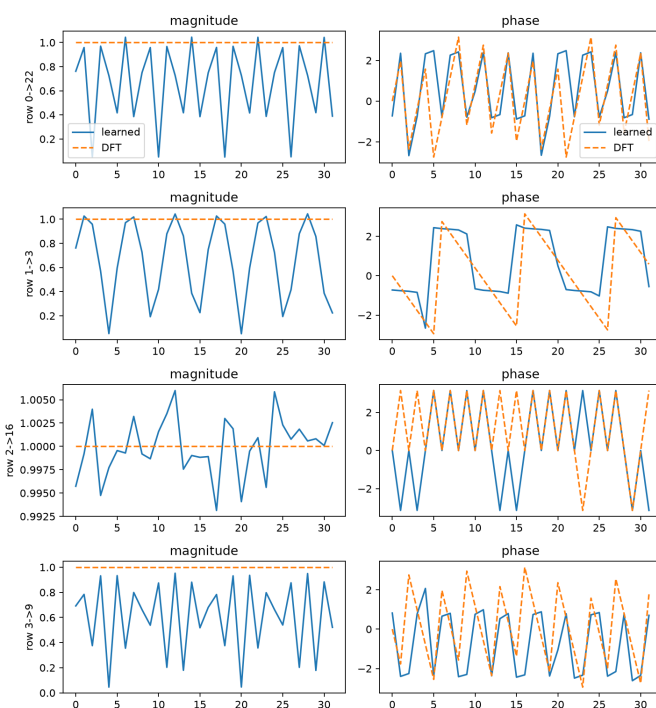
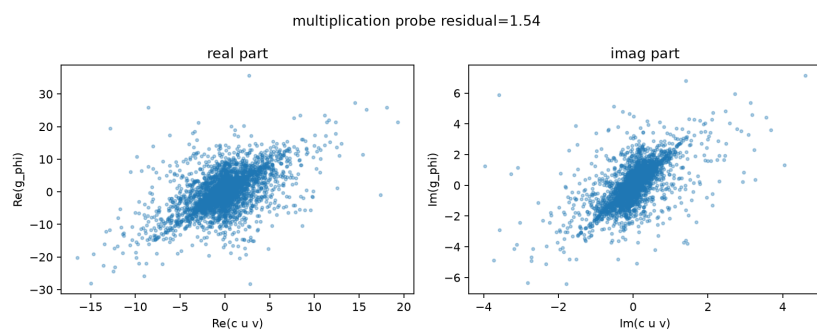
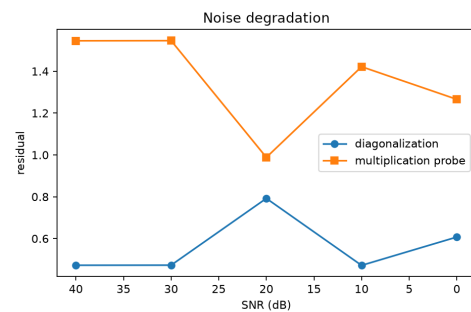


Figure 3: diag-only ablation: the basis is recovered, but the coordinate law is not trained.

(a) $|A|$ heatmap.

(b) Rows align poorly with the DFT.

(c) g_ϕ is not one multiplication law.

(d) Residuals versus SNR.

Figure 4: MSE-only ablation: without the diagonalization penalty, neither the basis nor the law is cleanly recovered.